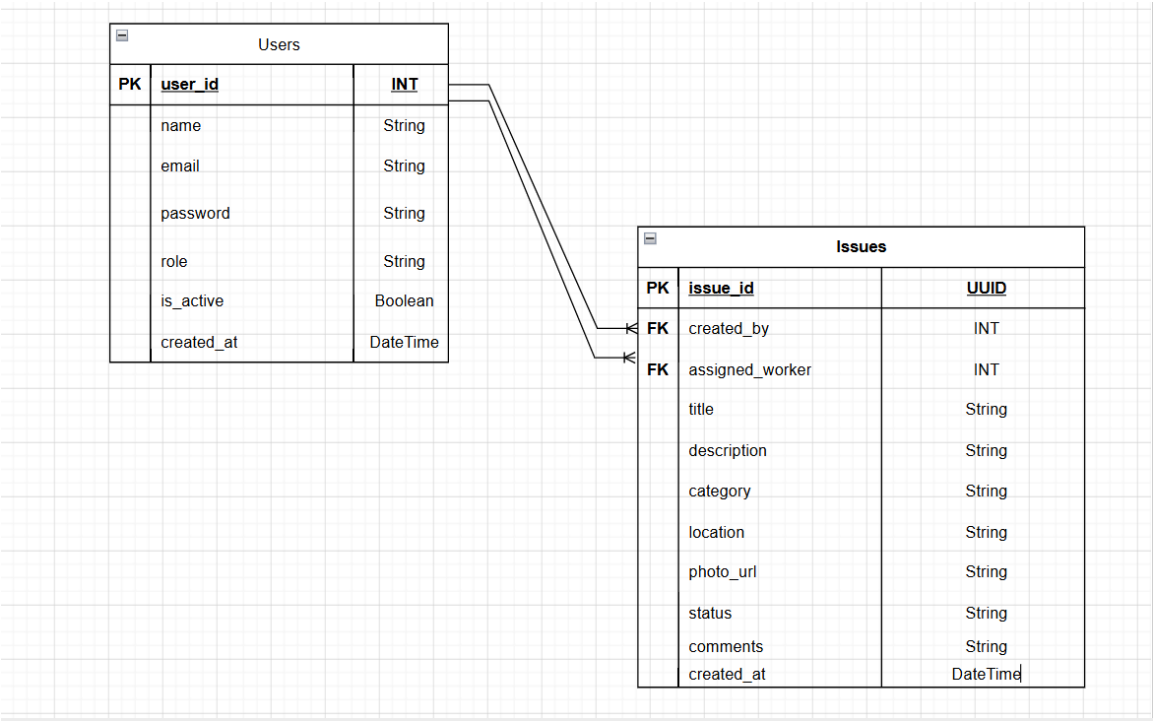
4.3 Database Schema Documentation

The database schema defines the structure of the CampusCare system database, including tables, relationships, primary keys, and foreign keys used to manage users, issues, and comments.

Entity Relationship Diagram (ERD)



Relationships

- One User creates many Issues (1:M)
- One Worker is assigned many Issues (1:M)

Users Table Documentation

Users Table:

Column Name	Data Type	Constraints
user_id	INT	Primary Key, Auto Increment
name	VARCHAR(100)	Not Null
email	VARCHAR(100)	Unique, Not Null
password	VARCHAR(255)	Not Null
role	VARCHAR(20)	Not Null
is_active	BOOLEAN	Default TRUE
created_at	DATETIME	Default Current Timestamp

Issues Table Documentation

Issues Table:

Column Name	Data Type	Constraints
issue_id	UUID	Primary Key
created_by	INT	Foreign Key
assigned_worker	INT	Foreign Key
title	VARCHAR(150)	Not Null
description	TEXT	Not Null
category	VARCHAR(50)	Not Null
location	VARCHAR(100)	Not Null
photo_url	TEXT	Nullable
status	VARCHAR(30)	Default 'Pending'
comments	TEXT	Nullable
created_at	DATETIME	Default Current Timestamp

Explain Foreign Keys

Foreign Keys:

Foreign Key References

created_by → Users(user_id)

assigned_worker → Users(user_id)

Sample Seed Data

Users Sample Data:

user_id	name	email	role
1	Ahmed Ali	ahmed@giu.edu.eg	Student
2	Sara Mohamed	sara@giu.edu.eg	Worker
3	Omar Hassan	omar@giu.edu.eg	Manager

Issues Sample Data:

Issue ID	Category	Status	Created By	Assigned Worker
I001	Electrical	Pending	U001	U002
I002	Cleanliness	In Progress	U001	U002

Conclusion

The database schema was designed to support issue reporting, assignment, status tracking, and communication between users, workers, and managers within the CampusCare system.